

Harmonizer Partwriting Guide

What the partwriting engine does today, rule by rule and weight by weight; how it measures against the September plan it was built from; and the six defects found in 2.30 and fixed in 2.31.

Short version. The greedy top-down filler of v2.4 is gone. Generate parts now reads the whole piece first (local key, secondary dominants, phrase ends and cadences, non-chord melody notes, suspensions), plans the bass line on its own, builds up to 16 candidate voicings for every melody note, and then picks the whole sequence at once with a search that scores three chords at a time. Almost every rule is a weighted cost; only a handful are absolute. Three style presets (Close harmony, Traditional (hymn), Open (wide)) change the weights.

The engine was written in v2.5, gained style presets in v2.12 and the look-back, cadence, suspension and local-key work in v2.18. Checking it for this guide turned up **six defects**, all fixed in **v2.31** (section 7): chords that dropped their third, a bass that could climb into the melody, no spacing check against the bass, 6/4 preferred to 6, a leading-tone exception aimed at the wrong note, and crossings around an inner-voice melody. Everything below describes

[harmonizer-2.31.html](#).

HARD never happens (filtered out, or a cost of 10^6)

STRONG a large cost; broken only when everything else is worse

SOFT a preference, weighed against the others

Costs are unitless and only their relative size matters. Pitches are sounding MIDI numbers (middle C = 60), so a tenor-clef note counts an octave below where it is written. Where a value differs by style it is given as *close* / *hymn* / *open*.

1. The pipeline

Everything lives in `generatePartsFromMelodyAndChords()` and the functions it calls. One press of Generate parts runs these steps over the whole song:

1. **Analyse** (`buildEvents`). One *event* per melody note, carrying the chord in force at the note's start. The local key is inferred, secondary dominants and cadences are marked, and each melody note is classified (section 2).
2. **Plan the bass** (`planBass`). The lowest non-melody stave is the bass. Its candidates are every chord tone at every octave within its range (± 2 semitones). A second-order Viterbi search picks the line using event costs, transition costs and three-note motion costs (section 3.2). Nothing about the upper voices is considered yet.
3. **Build voicings** (`beamVoicingsForEvent`). For each event, the staves are visited top-down in voice order, with the melody and bass already fixed. Each remaining voice tries every chord tone (plus 4–3 and 9–8 suspension tones where allowed) at every octave in range. Hard filters discard a candidate; costs accumulate. After each voice only the best **16** partial voicings are kept (a beam search).
4. **Search** (`viterbiUpperVoices` → `viterbi2`). A Viterbi over *pairs* of consecutive choices, so a cost can depend on three events in a row. Pair costs cover motion, parallels, hidden intervals, tendency tones and suspension preparation and resolution. Triple costs cover leap recovery and lockstep planing. The cheapest path through the whole piece wins.

5. **Write** (`materializeVoice`). Each voice gets the melody's rhythm and rests. Consecutive identical pitches are **tied**, not re-struck, so a voice holds through ornamental melody notes. Notes are spelled in the key signature in force where they sound. The melody's dynamics, hairpins, fermatas, articulations and breath marks are copied across.
6. **Report**. The status line counts cadences, suspensions and secondary dominants, and names any key areas heard outside the signature ("heard G major in mm. 9–16"). Melody notes that fit no reasonable voicing produce **chord suggestions**: buttons that apply a replacement chord and regenerate (section 2.6).

Locked notes. A hand-edited note on a generated stave locks itself; notes can also be locked explicitly. Before the search, locked notes are read back as fixed pitches. For the bass that means a single candidate; for an upper voice, a single option that skips the doubling cost. A re-run then repairs around them. Hand-built *Own rhythm* staves are left completely alone.

2. What the analysis hears

2.1 Local key (`inferLocalKeys`)

This is Krumhansl–Schmuckler key finding, measure by measure. It uses a five-measure window weighted $\frac{1}{4} \frac{1}{2} 1 \frac{1}{2} \frac{1}{4}$ over the melody, any hand-written own-rhythm staves, and the chords (root weighted 0.9, other chord tones 0.6).

BIAS	BONUS TO CORRELATION
Key signature in force	+0.30
Relative major/minor of the signature	+0.08
Same key as the previous measure	+0.15
Minimum run for a key other than the signature	3 measures, else reverted

The local key decides which chord counts as a dominant and which note is the leading tone. It never changes the notation: spelling follows the *written* key signature.

2.2 Dominant function and tendency tones (`chordTonesInfo`)

A chord is a dominant when its root is 5th of the local key and its quality is major or dominant-seventh. Its third is then the **leading tone**. The fourth tone of *any* seventh chord counts as the **chord seventh**. A vii° or $\text{vii}^\circ 7$, and a minor v , are not treated as dominants.

2.3 Secondary dominants (`markSecondaryDominants`)

A chord is heard as a secondary dominant when all of these hold:

- its root falls a fifth to the next chord;
- it is a dominant seventh, or a major triad that is not diatonic (I, IV and V in major; III, V, VI and VII in minor);
- the target is major, minor, $\text{maj}7$, $\text{m}7$ or 7, and is not the tonic.

The events under it are re-keyed to the target, so its third rises and its seventh falls like any $\text{V}7$.

2.4 Phrase ends and cadences (`markCadences`)

A note ends a phrase when it is:

- the last note;

- followed by at least half a beat of rest;
- at least 2 beats long and at least 3× the median note length; or
- the end of a tonic triad that arrives on a downbeat from V (or IV) and lasts 2 beats or more.

An end less than 4 beats after the previous one is a breath, not a cadence, and is dropped. For each phrase end, the event where the final chord arrives is marked as the cadence (2 for the piece's end). Every event under that chord is *in cadence*, and the event just before is *pre-cadence*.

Picardy third: at every cadence and on the last note, a melody singing the major third over a minor chord turns that chord major for the voicing search.

2.5 Melody notes

CLASS	HOW IT IS RECOGNISED	WHAT CHANGES
Chord tone	Pitch class is in the chord	Normal voicing; a generated voice a semitone (or major seventh) away costs <i>clash</i> .
Melody suspension	Same pitch as the previous note, across a chord change, not in the new chord, and falling 1-2 semitones to a chord tone next	Clash cost waived. Other voices avoid its resolution pitch (clash × 1.5).
Ornamental	Not in the chord, approached <i>or</i> left by step	Clash cost waived. There is no explicit "hold" rule: holding costs nothing in the motion table, and the chord usually has not changed, so voices hold.
Unexplained	Not in the chord and no step either side	Voiced like a chord tone. If it lasts a beat or more, it may produce a chord suggestion.

Passing, neighbour, appoggiatura, escape and anticipation notes are not told apart; all of them fall into "ornamental".

2.6 Chord suggestions (`collectChordSuggestions`)

These are offered for an unexplained melody note of a beat or longer. The engine tries every root (including the melody note itself and the written root) with every chord quality. It keeps the chord that contains the melody note and shares the most tones with the written chord, and merges repeats within 4 beats. Suggestions are proposed only, never applied without a click.

3. The rules as implemented

3.1 Ranges and tessitura

PART	STANDARD RANGE (SOUNDING)	PART	STANDARD RANGE (SOUNDING)
Descant	G4-E6	Tenor	C3-G4
Soprano	C4-A5	Tenor II	B $\flat$ 2-F4
Soprano II	B $\flat$ 3-G5	Baritone	G2-D4
Alto	G3-D5	Bass	E2-C4

PART	STANDARD RANGE (SOUNDING)	PART	STANDARD RANGE (SOUNDING)
Alto II	F3-C5	Bass II	D2-A3

RULE	STRENGTH	DETAIL
Search window	HARD	Standard range $\pm$ 2 semitones. Nothing outside it is ever a candidate.
Extended notes	SOFT	+2 for a note in those outer 2 semitones. There is no limit on how many in a row.
Tessitura	SOFT	The comfortable zone is 35–85% of the range. Cost is 0.15 per semitone below it and 0.3 per semitone above. The Open style moves the tenor zone down 3 and the bass zone down 5.
Range warnings	display	Settings → Range warnings tints notes outside the standard range (no $\pm$ 2 allowance). Same table.

3.2 The bass line (planned first)

RULE	STRENGTH	DETAIL
Chord tones only	HARD	Root, third, fifth or seventh of the written chord. The bass never takes a passing tone.
Position	SOFT	Root 0, third 1.5, seventh 4, fifth 6. A tonic chord moving to its dominant with $\hat{5}$ in the bass (the cadential 6/4) pays only 1.
Root at cadences	STRONG	Non-root bass: +4 on a cadence chord, +1.5 on a dominant just before a cadence, and +6 on the final chord.
Leading tone in bass	SOFT	+3 whenever the bass takes a dominant chord's leading tone. When a dominant resolves to its tonic, a bass leading tone must rise a semitone (bonus) or pay <i>unresolved</i> .
Bass motion	SOFT	Held 0.4 · step 1 · 4th/5th 1.5 · octave 1.5 · 3rd/6th 3 · tritone 6 · 7th 8 · beyond an octave 8 + 1 per semitone.
Against the melody	mixed	Parallel octaves HARD ; parallel fifths and hidden intervals by style (3.5). Contrary motion earns a bonus of 0.5 / 1.2 / 0.8.
Three-note shape	SOFT	The same leap-recovery rule as the upper voices (3.4), at half weight, with octave leaps exempt.
Below the melody	HARD	When the melody is on a higher stave, a bass above it costs 400.
Room for the other voices	STRONG	Melody to bass should span at least 6 + 3 semitones per voice between them (9 for SAB, an octave for SATB, 15 for five parts). Each semitone short costs 2.5.

3.3 Building each voicing

RULE	STRENGTH	DETAIL
Chord tones	HARD	Every generated note is a chord tone, a prepared suspension (3.7) or a locked note.

RULE	STRENGTH	DETAIL
No crossing	HARD	A voice may not sit above the voice above it. A unison is allowed but costs <i>unison</i> . No voice goes below the bass. Voices on staves above an inner-voice melody stay at or above it, and within the adjacent-gap limit of it.
Adjacent upper voices	HARD	At most 12 / 12 / 16 semitones apart (a voice and the one directly above it, melody included).
Upper spacing	SOFT	Close: +0.4 per semitone beyond a major sixth. Hymn: none. Open: +2 per semitone under a fifth, +0.5 per semitone beyond a ninth.
Gap above the bass	SOFT	The lowest upper voice pays 2 per semitone beyond a twelfth above the bass.
Low intervals	SOFT	Keyed to the lower note of two adjacent voices: 4th under C3 +1.5 · M3 under G2 +5 · m3 under C3 +5 · 2nd under G3 +6. Applied between every pair of adjacent voices, including the lowest upper voice and the bass.
Unisons	SOFT	+3 / +2.5 / +5 for each voice already on that exact pitch (melody and bass included).
Clash with melody	SOFT	+4 / +8 / +5 for a minor second or major seventh (any octave) against a melody note that is not ornamental.
Doubled leading tone or seventh	HARD	The second appearance of a dominant's leading tone, or of any chord seventh, is filtered out. The first appearance is fine.
Complete chords	SOFT	Coverage bonus −2.5 / −4 / −3 for a voice that supplies a chord factor nobody has yet. It is × 1.6 at cadences.
Missing third	STRONG	A finished voicing with no third anywhere (melody, bass and suspension resolutions included) pays +4, +10 on a cadence chord and +20 on the final chord. Sus chords are exempt.

Doubling costs (`doublingCost`). These are charged only to a voice that *doubles* a factor already present (in the melody, the bass or a voice above). A factor's first appearance is free.

CHORD / BASS	ROOT	THIRD	FIFTH	SEVENTH / OTHER
Triad, root in bass	0	2 / 5 / 3	1	3
Triad, third in bass	0	2 (min/dim) · 4	0.5	3
Triad, fifth in bass	6	6	0	6
Any seventh chord	0	5	1	3

3.4 Lines and motion

RULE	STRENGTH	DETAIL
Inner-voice motion	SOFT	Held 0 · 2nd 1 · 3rd 2 · 4th 4 · tritone 5 · 5th 3 · 6th 6 · 7th 9 · octave 3 · beyond 10 + 1 per semitone. Multiplied by 1 / 1.35 / 1.

RULE	STRENGTH	DETAIL
Leap recovery	SOFT	Three notes in one voice. After a leap of a 4th or more: continuing the same way costs 1.5 (by step) or 3.5 (by leap), and stepping back earns -0.8. Two leaps of a 3rd or more the same way that pass an octave cost +3; two that outline a tritone cost +1.2. All $\times 1 / 1.4 / 1$.
Lockstep planing	SOFT	All upper voices moving by the same interval twice running: 0 / 2.5 / 1.2.
Common tones	implicit	There is no rule by root motion. Common tones are kept because holding is free.
Forbidden melodic intervals	NONE	An augmented second costs the same as a minor third. Augmented fourths and sevenths are only expensive, not forbidden.

3.5 Parallels and hidden intervals

These are checked between **every pair** of melody, bass and generated voices, whenever both voices move in the same direction.

ERROR	STRENGTH	DETAIL
Parallel octaves and unisons	HARD	Octave or unison to octave or unison (compound intervals included), in every style. It is a 10^6 cost, not a filter, so locked notes can still force one. Two lines doubled at the octave are <i>not</i> exempt.
Parallel fifths	4 60 12	Perfect fifth to perfect fifth. Close tolerates them, Hymn effectively forbids them, Open discourages them. Unequal fifths are allowed.
Hidden 5ths/8ves	SOFT / STRONG	Checked for the melody-bass pair only. When the upper voice steps: 0 / 1.5 / 1. When it leaps: 7 / 14 / 9.
Contrary 5ths/8ves	NONE	Not checked.
Overlap, cross-relation	NONE	Not checked. Overlaps do occur, especially in Open style.

3.6 Tendency tones

These apply to generated voices only; the melody is never judged. The weights are multiplied by 1.6 on cadence chords.

TONE	WANTED	BONUS / COST	NOTES
Leading tone of a dominant resolving to its tonic	Up a semitone	-3 / -4 / -3 else +5 / +10 / +6	A half bonus is given for falling a major third to $\hat{5}$, the inner-voice exception.
Chord seventh (any seventh chord)	Down by step	$0.6 \times \text{bonus}$ else $0.7 \times \text{cost}$	No cost if the next chord still contains it, so it may be held.
Secondary leading tones	As above	as above	Via the re-keying in 2.3.
Cadential 6/4, deceptive V-vi	n/a	none	Not recognised.

3.7 Suspensions (v2.18)

- **Types:** 4–3 (the fourth above the root, resolving to the third) and 9–8 (the ninth, resolving to the root). They are written only over triads.
- **When:** the chord changes on this melody note and lasts into the next; the melody note is a chord tone at least half a beat long; the suspended pitch class was in the previous chord (prepared).
- **Not when:** the melody has the suspended note or its resolution now, or sings the resolution next. A 4–3 is also skipped when the bass has the third.
- **Preparation and resolution:** the note must be held from the event before and move to its resolution at the event after. Breaking either costs **60**, which is effectively hard.
- **Weights:** bonus of -0.8 / -0.3 / -0.5 , plus -2.2 on or just before a cadence. The suspension counts as the chord factor it resolves to. Other voices pay $+6$ to sound its resolution pitch class at the same time.

4. Style presets

The preset is chosen from the menu beside Generate parts and saved with the song. Older songs, or songs with an unknown style, open as Close harmony.

SETTING	CLOSE HARMONY	TRADITIONAL (HYMN)	OPEN (WIDE)
Parallel fifths	4	60	12
Hidden 5th/8ve (step / leap)	0 / 7	1.5 / 14	1 / 9
Unison	3	2.5	5
Clash with melody	4	8	5
Tendency resolved / unresolved	-3 / 5	-4 / 10	-3 / 6
Coverage bonus	-2.5	-4	-3
Third doubling (root position)	2	5	3
Max adjacent upper gap	12	12	16
Spacing curve	pull toward close	none	push toward a 5th–10th
Tessitura shift	none	none	tenor -3 , bass -5
Motion weight	1	1.35	1
Contrary bass bonus	0.5	1.2	0.8
Leap recovery weight	1	1.4	1
Lockstep planing	0	2.5	1.2
Suspension bonus	-0.8	-0.3	-0.5

Chord vocabulary does not depend on the style. The engine voices whatever is written: maj, m, dim, aug, 7, maj7, m7, dim7, sus2, sus4.

5. Parts, divisi and melody placement

- **Up to 8 staves.** The lowest non-melody staff gets bass treatment. *Every* other staff, including a second low part, is voiced as an upper voice under the same rules. There is no per-voice-count doubling table and no "low end wide" rule.
- **Divisi.** Generate parts writes one line per staff. Two-note divisi stems come only from hand entry, Combine, or import. The *Mark parallel fifths & octaves* display treats a divisi as its own line and ignores two staves singing identical notes. Generation itself has no octave-doubling exemption.
- **Melody on any staff.** The melody can be on any staff, and the other voices are placed around it. Voices above it stay above it, and the bass stays below it. A melody on the *lowest* staff is the one case not handled: the "bass" is then planned above it as if it were an upper voice.
- **Eligible staves.** A staff can be filled if it is empty, is an empty own-rhythm staff, or holds a previous Generate parts result. Changing the melody, the chords, or a staff's Part or Clef clears stale harmony.
- **Rhythm.** Generated voices share the melody's rhythm and rests, tied where the pitch holds. Chords are sampled once per melody note, at its start. *A chord change in the middle of a long melody note is not heard.*

6. Against the September plan

The v2.4-era guide ended with an integrated rule set (its section 4) and a proposed algorithm (its section 5). This table shows where each part of that plan stands.

PLANNED	STATUS	NOW
Score whole voicings; search the phrase	DONE	Beam (16) plus a second-order Viterbi over the whole piece, not per phrase.
Plan the bass first	DONE	Its own Viterbi pass with cadence and root-motion costs.
Inversions used well	PARTIAL	First inversion preferred; 6/4 kept for the cadential 6/4. Passing and pedal 6/4 are not recognised.
Ranges and tessitura	DONE	± 2 extension with no one-note limit. There is no top-of-range reserve for phrase peaks.
Spacing and low-interval table	PARTIAL	Applied between every adjacent pair, bass included. No acoustic-pyramid reward.
Doubling table	DONE	Charged on real doubles only, plus a missing-third cost.
Smooth lines, leap recovery	DONE	Graded motion costs and three-note recovery.
Forbidden melodic intervals	MISSING	Only graded costs by size.
Progression-aware common tones	PARTIAL	Only implicit (holding is free).

PLANNED	STATUS	NOW
Parallel 5ths/8ves, every pair	DONE	Contrary-motion 5ths/8ves are not checked, and octave doublings are not exempt.
Hidden 5ths/8ves, outer voices	PARTIAL	Checked for melody versus bass, not the highest versus lowest sounding voice.
Crossing; overlap; cross-relation	PARTIAL	Crossing prevented for every stave, above or below the melody. Overlap and cross-relation are not checked.
Leading-tone and seventh resolution	DONE	Soft, stronger at cadences, secondary dominants included, with the $\hat{7} \rightarrow \hat{5}$ inner-voice exception.
Classify melody notes	PARTIAL	Chord tone, ornamental, melody suspension, or unexplained. No finer types.
Cadences and phrases	PARTIAL	Phrase detection, root-position bass, complete chords, stronger resolution, Picardy third. No cadential 6/4, deceptive-cadence doubling or phrase-peak rule.
Suspensions in generated voices	DONE	Prepared 4-3 and 9-8 (not in the plan's list; added in v2.18).
Local key and modulation	DONE	Key inference plus secondary dominants (also beyond the plan).
Style profiles	DONE	Close / Hymn / Open instead of the planned Chorale / Close / Jazz-pop. Vocabulary is not style-dependent.
Five to eight parts	PARTIAL	Voices are filled, but the per-count doubling, split-order and low-end rules are missing.
Split events at chord changes	MISSING	One chord per melody note.
Locked notes	DONE	Fixed pitches in the search.
Region regeneration	MISSING	It is whole piece or nothing; locking is the workaround.
Linter / quality report	PARTIAL	Status-line counts, parallel marks and range warnings. No rule-by-rule list of what was paid for, and where.

7. Defects found and fixed in 2.31

These were found by running v2.30 through the `tests/dom-stub.js` harness. The test tune was a 14-chord SATB chorale in C, with every melody note a chord tone (G G F E D E G A G G F E D C over C C Dm C G C C F C C G7 C G C), generated in each style. The same chorale is now a regression test in `logic.test.js` (13 checks, which fail on 2.30 and pass on 2.31). The full suite passes: 289 logic and 102 browser checks.

DEFECT IN 2.30	WHAT WAS HEARD	FIX IN 2.31
7.1 Chords lose their third	The doubling cost was charged on a factor's <i>first</i> appearance. Chords without a third: 5/14 in Hymn, 3/14 in Close, 2/14 in Open. A G7 was voiced F D G G, with no leading tone, in every style.	Doubling costs apply to real doubles only, plus the missing-third cost (3.3). Now 0/14 in every style, and the G7 keeps its B.

DEFECT IN 2.30	WHAT WAS HEARD	FIX IN 2.31
7.2 Bass climbs into the melody	Hymn's final chord came out C4 C4 C4 C4. Other chords had tenor and bass in unison a fifth under the melody.	Room rule for the bass (3.2). The final chord is now C4 G3 E3 C3 in every style.
7.3 No check against the bass	The tenor-bass gap and low-interval costs never ran (their condition was effectively never true).	Both are applied between the lowest upper voice and the bass.
7.4 6/4 cheaper than 6	C over G straight after the G7 (E4 C4 G3 G3) in Close and Hymn.	Third 1.5, fifth 6, cadential 6/4 1. That C chord is now in root position.
7.5 Leading-tone exception	A $-7 / +5$ test rewarded $\hat{7} \rightarrow \hat{3}$.	Now tests -4 ($\hat{7} \rightarrow \hat{5}$). Close now writes the alto's B3 $\rightarrow$ G3 at the final cadence.
7.6 Inner-voice melody	With the tune in the tenor, the bass sat above it at 9 of 26 notes.	A floor at the melody for staves above it; the bass is kept below it. Crossings: 0.

Still open

- Contrary 5ths/8ves, voice overlap and cross-relations are not checked. Octave doublings are not exempt from the parallel-octave rule.
- Melodic augmented seconds are not forbidden. Passing and pedal 6/4 are not recognised, and neither is a deceptive cadence.
- One chord per melody note: a chord change in the middle of a long note is not heard.
- In Hymn style the last V before a cadence sometimes comes out as V6 (bass on the leading tone, resolving up). That is correct, but a root-position V would be more usual for a perfect authentic cadence.
- There is no rule-by-rule quality report, and no regeneration of just a region.

Minor

- Parallel octaves are a 10^6 cost rather than a filter. If locked notes force one on every path, the search still returns a result, and nothing reports that it happened.

8. Code map

```
HARMONY_STYLES / STYLE_CLOSE, STYLE_HYMN, STYLE_OPEN  
VOICE_RANGE_SOUNDING, ROLE_RANK  
chordTonesInfo, withPicardyThird  
inferLocalKeys, buildEvents, markSecondaryDominants,  
markCadences  
collectLockedAnchors  
tessituraCost, styleRange, doublingCost,  
lowIntervalCost, bassGapCost  
voiceMotionCost, bassMotionCost, secondOrderMotion  
pairTransitionCost, tendencyTransitionCost,  
transitionCost  
planBass, bassEventCost, bassTransitionCost  
beamVoicingsForEvent, viterbiUpperVoices, viterbi2  
materializeVoice  
collectChordSuggestions  
findParallels, rangeStatus
```

weights (section 4)
ranges and voice order
chord factors, tendency tones

analysis (section 2)
locked notes → fixed pitches

per-note costs
line costs

between-chord costs
bass pass
voicing search
write-out with ties and marks
chord proposals
display-only checks (Settings)

Source: [harmonizer-2.31.html](https://github.com/hammerlab/harmonizer-2.31.html). Checked 25 September 2026.